Boreddy Sri Sai Abhinav Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.30
Sri Chaitanya Junior kalashala	Class XII (Telangana State Board of Intermediate Education)	2024	98.40%
Dr KKR's Gowtham Concept School	Class X (Telangana State Board of Secondary Education)	2022	9.80%

SCHOLASTIC ACHIEVEMENTS

- Secured **State Top 1% in NSEP examination** and **National Top 1% in NSEC examination** and qualified and appeared for the **INChO(Indian National Chemistry Olympiad)** in Class XII
- Campus Finalist in the Optiver presented **THE INTEGRAL CUP** and qualified for the **Round-2 in IIT Hyderabad**
- Secured **5th position in RESCON 2025** conducted by EnPower IIT Bombay for the submission of **Effect of Kernel sizes of the Robustness of Neural Networks**

WORK EXPERIENCE

Theoretical Computer Science Summer Internship in (IMSc-Chennai)

The Institute of Mathematical Sciences : *May 26- Jun26*

- Worked on Topics like Random K-out graphs and expected number of their inter component edges.
- Got an exposure to various theoretical computer science domains like algorithms,lower bounds, computation, complexity theory,computability, exact expoenential algorithms.

PROJECTS

Convex Clustering and Mixture Modelling:

- Developed and analyzed convex clustering frameworks with structured weighting and 1-fusion penalties, extending traditional mixture modeling approaches with stronger optimization and recovery guarantees

Effect of Kernel Size on Robustness of Neural Networks:

- Conducted empirical analysis of the impact of convolution kernel sizes on CNN robustness against memory bit-flip errors using ResNet-18 and CIFAR-10.Identified optimal kernel-size trade-offs that improve fault tolerance while balancing overheads in some usecases.

Locally Binary Functions Analysis:

- Established graph-theoretic connections between locally binary functions and independence numbers of Hamming-distance graphs to characterize function expressiveness.Derived bounds on expressiveness and entropy, with applications in coding theory, communication systems, and function-correcting codes.

Infinite Arithmetic Precision Library for Java

- Built a Java library for arbitrary-precision integer and floating-point arithmetic from scratch, supporting high-precision computations beyond native data type limits,Implemented core arithmetic operations, input validation, exception

handling.

RISC-V Simulator for Custom ISA Extensions

- Designed and implemented a custom RISC-V simulator supporting SIMD operations, low-precision floating-point formats (BFloat16/FP16/MSFP16), ECC mechanisms, and a Quantum-Inspired ALU

xv6 Operating System Extensions:

- Enhanced xv6 with custom system calls, SC-MLFQ scheduling, virtual memory and page replacement, disk-backed swapping, RAID 0/1/5 storage, FCFS/SSTF disk scheduling, and kernel performance monitoring.

SKILLS

Programming Languages: C++,Python,C,SQL,HTML,CSS,Javascript,Bash scripting,RISC-V (assembly)

Tools and Libraries: Numpy,Pandas,Matplotlib (basics),Verilog(basics)

RELEVANT COURSES

- **Theory CS Courses:** Discrete Mathematics, Data Structures & Applications, Design and Analysis of Algorithms, Theory of Computation
- **Mathematics Courses:** Concentration Inequalities, Convex Optimisation, Differential Equations, Discrete Mathematics, Intro Group Theory, Linear Algebra, Probability & Random Variables, Random Processes, Statistics, Transform Techniques
- **Programming and Systems Courses:** Compilers-1, Computer Architecture, DBMS I, DBMS-II, Intro to Computing, Operating Systems I,Operating systems-II, Software Development Fundamentals, Digital Circuits

POSITIONS OF RESPONSIBILITY

- Did **Teaching Assistance(TA ship)** for the course **Discrete Mathematics for CS (CS1010)** under prof Rogers Mathew - *Jul to Nov 2025*

- Did **Teaching Assistance(TA ship)** for the course **Probability and Random Variables (AI1110)** under prof Anjana AM - *Jan to Apr 2026*

- Class Representative of CSE-28 for academic year 2026-2027